

Cambridge (CIE) IGCSE Additional Maths



Your notes

Simultaneous Equations

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Your notes

Quadratic Simultaneous Equations

Quadratic Simultaneous Equations

What are quadratic simultaneous equations?

- When there are two unknowns (say x and y) in a problem, we need two equations to be able to find them both: these are called **simultaneous equations**
- If there is an x^2 or y^2 or xy in one of the equations then they are **quadratic** (or **non-linear**) simultaneous equations

How do I solve quadratic simultaneous equations?

- Use the method of substitution
 - **Substitute** the **linear** equation, $y = \dots$ (or $x = \dots$), into the quadratic equation
 - Do not try to substitute the quadratic equation into the linear equation
- Solve $x^2 + y^2 = 25$ and $y - 2x = 5$
 - **Rearrange** the **linear** equation into $y = 2x + 5$
 - **Substitute** this into the quadratic equation, replacing all y 's with $(2x + 5)$ in **brackets**
 - $x^2 + (2x + 5)^2 = 25$
 - Expand and **solve** this **quadratic** equation ($x = 0$ and $x = -4$)
 - Substitute **each** value of x into the **linear** equation, $y = 2x + 5$, to get their value of y
 - Present your solutions in a way that makes it obvious which x belongs to which y
 - $x = 0, y = 5$ or $x = -4, y = -3$
- **Check** your final solutions satisfy both equations

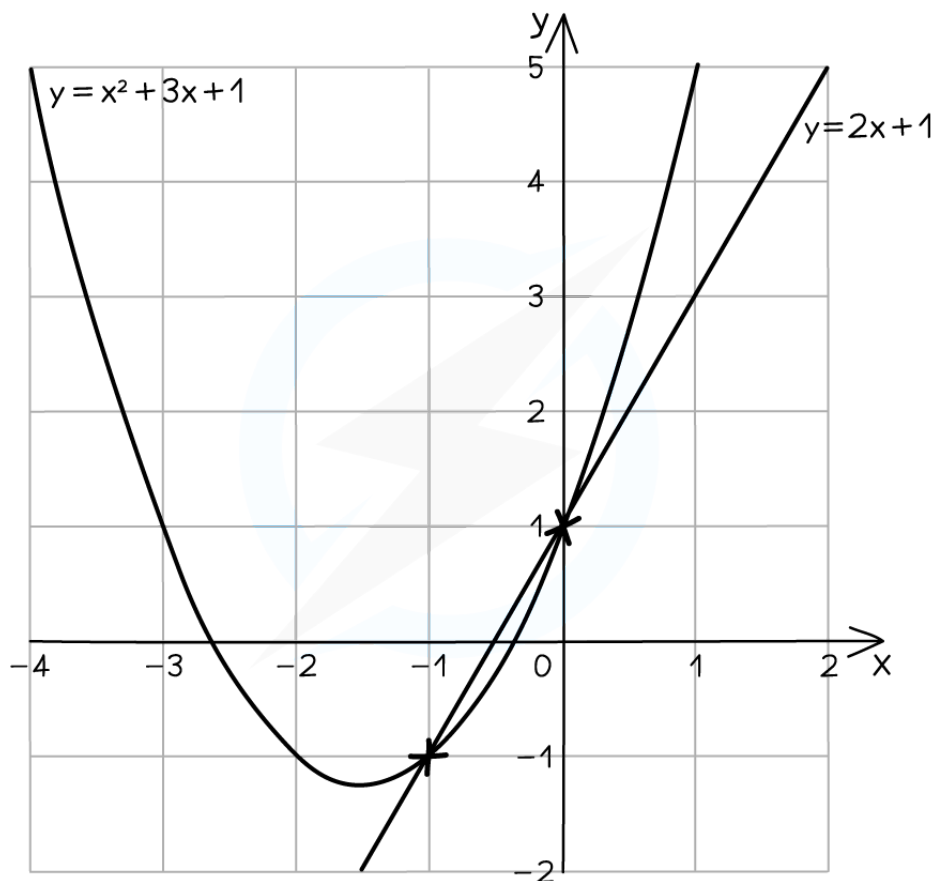
How do you use graphs to solve quadratic simultaneous equations?

- **Plot** both equations on the same set of axes
 - to do this, you can use a table of values (or, for straight lines, rearrange into $y = mx + c$ if it helps)
- Find where the lines **intersect** (cross over)

- The **x** and **y solutions** to the simultaneous equations are the **x** and **y coordinates** of the point of **intersection**
- e.g. to solve $y = x^2 + 3x + 1$ and $y = 2x + 1$ simultaneously, first plot them both (see graph)
 - find the points of intersection, $(-1, -1)$ and $(0, 1)$
 - the solutions are $x = -1$ and $y = -1$ or $x = 0$ and $y = 1$



Your notes



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Examiner Tips and Tricks

- If the resulting quadratic has a **repeated root** then the line is a **tangent** to the curve (as there is only 1 solution)



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- If the resulting quadratic has **no roots** then the line does not intersect with the curve (as there are 0 solutions) – or you have made a mistake!
- When giving your final answer, make sure you indicate which x and y values go together
 - If you don't make this clear you can lose marks for an otherwise correct answer



Worked Example

Solve the equations

$$x^2 + y^2 = 36$$

$$x = 2y + 6$$

Number the equations.

$$x^2 + y^2 = 36 \quad (1)$$

$$x = 2y + 6 \quad (2)$$

There is one quadratic equation and one linear equation so this must be done by substitution.

Equation (2) is equal to x so this can be eliminated by substituting it into the x part for equation (1).

Substitute $x = 2y + 6$ into equation (1).

$$(2y + 6)^2 + y^2 = 36$$

[1]

Expand the brackets, remember that a bracket squared should be treated the same as double brackets.

$$(2y + 6)(2y + 6) + y^2 = 36$$

$$4y^2 + 6(2y) + 6(2y) + 6^2 + y^2 = 36$$

[1]

Simplify.

$$4y^2 + 12y + 12y + 36 + y^2 = 36$$

$$5y^2 + 24y + 36 = 36$$

[1]

Rearrange to form a quadratic equation that is equal to zero.



Your notes

$$5y^2 + 24y + 36 - 36 = 0$$
$$5y^2 + 24y = 0$$

The question does not give a specified degree of accuracy, so this can be factorised. Take out the common factor of y .

$$y(5y + 24) = 0$$

Solve to find the values of y .

Let each factor be equal to 0 and solve.

$$y_1 = 0$$

$$5y_2 + 24 = 0 \Rightarrow y_2 = -\frac{24}{5} = -4.8$$

[1]

Substitute the values of y into one of the equations (the linear equation is easier) to find the values of x .

$$x_1 = 2(0) + 6 = 6 \quad x_2 = 2\left(-\frac{24}{5}\right) + 6 = -9.6 + 6$$

$$x_1 = 6, y_1 = 0$$

$$x_2 = -3.6, y_2 = -4.8 \quad [1]$$



Your notes

Linear Simultaneous Equations

Elimination Method

What are simultaneous linear equations?

- When there are two unknowns (say x and y) in a problem, we need two equations to be able to find them both: these are called **simultaneous equations**
 - you solve **two** equations to find **two** unknowns, x and y
 - for example, $3x + 2y = 11$ and $2x - y = 5$
 - the solutions are $x = 3$ and $y = 1$
- If they just have x and y in them (no x^2 or y^2 or xy etc) then they are **linear** simultaneous equations
- Solving a **pair of simultaneous equations** means finding **pairs of values** that make **both equations true** at the same time
- A linear equation in two unknowns will produce a straight line if you graph it... linear = line
- A pair of simultaneous equations will produce lines that will cross each other (if there is a solution!)

How do I use elimination to solve simultaneous linear equations?

- "Elimination" completely removes one of the variables, x or y
- Begin by multiplying one (or both) of the equations by a constant (or constants) to get the numbers in front of **one** of the unknowns to match
 - For example to eliminate the x 's from $3x + 2y = 11$ and $2x - y = 5$
 - Multiply every term in the first equation by 2 $6x + 4y = 22$
 - Multiply every term in the second equation by 3 $6x - 3y = 15$
- If the matching numbers have the **same** sign, then **subtract** one equation from the other
- If the matching numbers have **different** signs then **add** the equations together
 - Subtract the second result from the first to eliminate the $6x$'s
 - $4y - (-3y) = 22 - 15$
 - $7y = 7$
- Solve the new equation to find the value of one of the unknowns



Your notes

- Solve to find y
 - $y = 1$
- Substitute the value into one of the original equations and solve to find the value of the other unknown
 - Substitute $y = 1$ back into either original equation
 - $3x + 2(1) = 11$
 - $x = 3$
 - Alternatively, to eliminate the y 's from $3x + 2y = 11$ and $2x - y = 5$
 - Multiply every term in the second equation by 2
 - $4x - 2y = 10$
 - **Add** this result to the first equation to eliminate the $2y$'s (as $2y + (-2y) = 0$)
 - The process then continues as above
- **Check** your final solutions satisfy both equations
 - $3(3) + 2(1) = 11$ and $2(3) - (1) = 5$

How do I solve linear simultaneous equations from worded contexts?



Your notes

e.g. AT A FIREWORKS STALL A CUSTOMER PAYS £9 FOR SIX BANGERS AND TWELVE SPARKLERS. ANOTHER CUSTOMER BUYS NINE BANGERS AND TEN SPARKLERS FOR £12.30. FIND THE COST OF 5 BANGERS AND 15 SPARKLERS.

TWO 'THINGS' – B's AND S's
TWO UNKNOWN'S – COST OF B's AND S's
TWO PIECES OF INFO – TWO EQUATIONS
SIMULTANEOUS EQUATIONS

NO 'NICE' LINK FROM 6 OR 9 TO 5,
NOR FROM 12 OR 10 TO 15, SO FIND
COST OF 1B AND 1S

NOTICE HOW MUCH WORK AND THINKING SHOULD
HAPPEN IN YOUR HEAD BEFORE ATTEMPTING A SOLUTION!

LET B BE THE PRICE OF BANGERS, IN POUNDS.
LET S BE THE PRICE OF SPARKLERS, IN POUNDS.

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$$6B + 12S = 9 \quad (1)$$

$$9B + 10S = 12.3 \quad (2)$$

$$(1) \times 3: 18B + 36S = 27 \quad (3)$$

$$(2) \times 2: 18B + 20S = 24.6 \quad (4)$$

$$(3) - (4): 16S = 2.4$$

$$S = \frac{2.4}{16}$$

$$S = 0.15$$

$$\text{SUB. IN } (3): 18B + 36 \times 0.15 = 27$$

$$18B = 21.6$$

$$B = \frac{21.6}{18}$$

$$B = 1.2$$

$$\begin{aligned} \text{SO, } 5B + 15S &= 5 \times 1.2 + 15 \times 0.15 \\ &= 8.25 \end{aligned}$$

FIVE BANGERS AND FIFTEEN SPARKLERS
WILL COST £8.25

A GOOD ALTERNATIVE QUESTION HERE TO TEST
INTERPRETATION OF YOUR ANSWER WOULD BE
TO FIND THE CHANGE FROM £10. (£1.75!)

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Examiner Tips and Tricks

- Don't skip the checking step (it only takes a few seconds) – there are many places to go wrong when solving simultaneous equations!
- Mishandling minus signs is probably the single biggest cause of student error in simultaneous equations questions



Worked Example



Your notes



Your notes



Solve the simultaneous equations:

$$-2x + 4y = 5$$

$$4x - 5y = -7$$

LABEL THE
EQUATIONS

$$-2x + 4y = 5 \quad \textcircled{1}$$

$$4x - 5y = -7 \quad \textcircled{2}$$

STEP 1: $[2 \times \textcircled{1}] \quad -4x + 8y = 10$
 $[\textcircled{2}] \quad 4x - 5y = -7$

WE ONLY HAVE TO
MULTIPLY ONE OF THE
EQUATIONS HERE

STEP 2:
$$\begin{array}{r} -4x + 8y = 10 \\ + (4x - 5y = -7) \\ \hline 3y = 3 \end{array}$$

THE SIGNS IN FRONT OF
THE 4's ARE DIFFERENT,
SO ADD

STEP 3: $y = 3 \div 3 = 1$

STEP 4:
$$\begin{array}{l} 4x - 5(1) = -7 \\ 4x = -2 \\ x = -0.5 \end{array}$$

USING EQUATION $\textcircled{2}$

STEP 5:
$$\begin{array}{l} -2(-0.5) + 4(1) = 5 \\ 1 + 4 = 5 \\ 5 = 5 \checkmark \end{array}$$

USING EQUATION $\textcircled{1}$

$$x = -0.5 \quad y = 1$$

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Substitution Method

How do I use substitution to solve simultaneous linear equations?

- "Substitution" means substituting one equation into the other
- Rearrange one of the equations to make one of the unknowns the subject
 - To solve $3x + 2y = 11$ and $2x - y = 5$ by substitution
 - Rearrange one of the equation into $y = \dots$ (or $x = \dots$)
 - For example, the second equation becomes $y = 2x - 5$
- Substitute the expression found for x or y into the equation not used to rearrange
 - Replace all y 's with $2x - 5$ in brackets
 - $3x + 2(2x - 5) = 11$
- Solve the new equation to find the value of one of the unknowns
 - Solve this equation to find x
 - $x = 3$
- Substitute the value found for x or y into the rearranged equation from to find the value of the other unknown
 - Substitute $x = 3$ into $y = 2x - 5$ to find y
 - $y = 2(3) - 5$
 - $y = 1$
- **Check** your final solutions satisfy both equations
 - $3(3) + 2(1) = 11$ and $2(3) - (1) = 5$

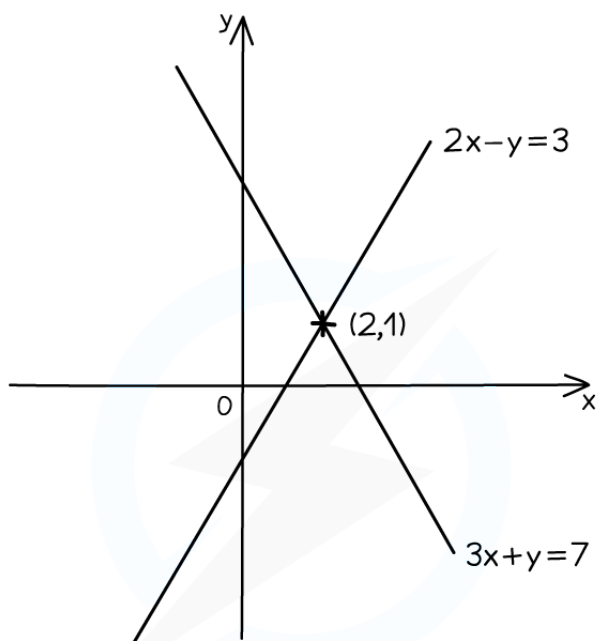
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- e.g. to solve $2x - y = 3$ and $3x + y = 4$ simultaneously, first plot them both (see graph)
 - find the point of intersection, (2, 1)
 - the solution is $x = 2$ and $y = 1$



Your notes



- LINES INTERSECT AT (2,1)
- SOLVING $2x - y = 3$ AND $3x + y = 7$ SIMULTANEOUSLY IS $x = 2$, $y = 1$

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Examiner Tips and Tricks

- Although elimination will always work to solve simultaneous linear equations, sometimes substitution can be easier and quicker

- Knowing both methods can help you a lot in the exam (plus you will need substitution to solve quadratic simultaneous equations)



Worked Example



Your notes



Your notes



Solve the simultaneous equations:

$$3x + 5y = -9$$

$$y = 1 - 9x$$

LABEL THE
EQUATIONS

$$3x + 5y = -9 \quad \textcircled{1}$$

$$y = 1 - 9x \quad \textcircled{2}$$

y IS ALREADY THE SUBJECT
OF EQUATION ②, SO WE
CAN SKIP STEP 1

STEP 2: $3x + 5(1 - 9x) = -9$
 $3x + 5 - 45x = -9$
 $-42x = -14$

SUBSTITUTE EQUATION ②
INTO EQUATION ①

STEP 3: $x = -14 \div (-42) = \frac{1}{3}$

STEP 4: $y = 1 - 9(\frac{1}{3}) = 1 - 3 = -2$

USING EQUATION ②

STEP 5: $3(\frac{1}{3}) + 5(-2) = -9$
 $1 - 10 = -9$
 $-9 = -9 \checkmark$

USING EQUATION ①

$$x = \frac{1}{3} \quad y = -2$$

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